

Modelado de Sistemas 17 Feb - 2026

$$V(s) = \left[ (R_1 + \frac{1}{Cs} + R_2) \frac{CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} - \left( \frac{1}{Cs} + R_2 \right) \right] I_1$$

$$Z_x = \frac{1}{Cs} + R_2 = \frac{CR_2s + 1}{Cs}$$

$$V(s) = \left[ (R_1 + Z_x) \cdot \frac{CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} - Z_x \right] I_2(s)$$

$$(R_1 + Z_x) \cdot \frac{\text{Num}}{\text{Den}} = R_1 \frac{CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} + \left( \frac{CR_2s + 1}{Cs} \right) \left( \frac{CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} \right)$$

$$R_1 \frac{CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} + \frac{CLs^2 + C(R_2 + R_3)s + 1}{Cs}$$

$$\frac{R_1 CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} + \frac{CLs^2 + CR_2s + CR_3s + 1 - (CR_2s + 1)}{Cs}$$

$$\frac{CLs^2 + CR_2s}{Cs} = \frac{Cs(Ls + R_2)}{Cs} = Ls + R_2$$

$$V(s) = \left[ \frac{R_1 CLs^2 + C(R_2 + R_3)s + 1}{CR_2s + 1} + (Ls + R_2)(CR_2s + 1) \right] I_2(s)$$

$$V(s) = \left[ \frac{CL(R_1 + R_2)s^2 + C(R_1R_2 + R_1R_3 + R_2R_3 + L)s + (R_1 + R_2)}{CR_2s + 1} \right] I_2(s)$$

$$\frac{1}{V(s)} = \frac{CR_2R_3s + R_3}{CL(R_1 + R_2)s^2 + C(R_1R_2 + R_1R_3 + R_2R_3 + L)s + (R_1 + R_2)}$$



Método de Sistemas

17-Set-2027

$$\frac{I_1(s)}{cs} - \frac{I_2(s)}{cs} + R_2 I_1(s) - R_2 I_2(s) = Ls I_2(s) + R_3 I_2(s)$$

$$\left(\frac{1}{cs} + R_2\right) I_1(s) = (Ls + R_3 + R_2 + \frac{1}{cs}) I_2(s)$$

$$\frac{1}{cs} + R_2 = \frac{1 + R_2 cs}{cs}$$

$$Ls + R_3 + R_2 + \frac{1}{cs} = \frac{(Ls)(cs) + (R_3)(cs) + (R_2)(cs) + 1}{cs}$$

$$\frac{Lcs^2 + (R_3 + R_2)cs + 1}{cs}$$

$$\frac{R_2 cs + 1}{cs} I_1(s) = \frac{Lcs^2 + (R_2 + R_3)cs + 1}{cs} I_2(s)$$

$$I_1(s) = \frac{Lcs^2 + (R_2 + R_3)cs + 1}{cs} \cdot \frac{cs}{R_2 cs + 1} I_2(s)$$

$$I_1(s) = \frac{Lcs^2 + (R_2 + R_3)cs + 1}{R_2 cs + 1} I_2(s)$$

$$V_1(s) = (R_1 + \frac{1}{cs} + R_2) I_1(s) - (\frac{1}{cs} + R_2) I_2(s)$$

$$V_1(s) = (R_1 + \frac{1}{cs} + R_2) \frac{Lcs^2 + (R_2 + R_3)cs + 1}{R_2 cs + 1} I_2(s) - (\frac{1}{cs} + R_2) I_2(s)$$



Scanned with CamScanner



# Modelado de Sistemas

17-abr-2026

Se sustituye en la ecuación de la primera malla (del voltaje de entrada)

$$V_e(s) = (Ls I_1(s) + R_1 I_1(s) + R_2 [I_1(s) - I_2(s)])$$

$$V_e(s) = Ls I_1(s) + R_1 I_1(s) + R_2 I_1(s) - R_2 I_2(s)$$

$$V_e(s) = (Ls + R_1 + R_2) I_1(s) - R_2 I_2(s)$$

$$V_e(s) = (Ls + R_1 + R_2) \frac{C(R_2 + R_3)s + 1}{CR_2s} I_2(s) - R_2 I_2(s)$$

$$V_e(s) = \left( \frac{(Ls + R_1 + R_2)(C(R_2 + R_3)s + 1)}{CR_2s} - R_2 \right) I_2(s)$$

$$\frac{(Ls + R_1 + R_2) \cdot C(R_2 + R_3)}{(Ls + R_1 + R_2) \cdot 1}$$

$$R_2 = \frac{R_2 \cdot (CR_2s)}{CR_2s} = \frac{CR_2^2s}{CR_2s}$$

$$V_e(s) = \left( \frac{(Ls + R_1 + R_2)(C(R_2 + R_3)s + 1) - CR_2^2s}{CR_2s} \right) I_2(s)$$

$$V_e(s) = \frac{LC(R_2 + R_3)s^2 + [L + C(R_1R_2 + R_1R_3 + R_2R_3)]s + (R_1 + R_2)}{CR_2s} I_2(s)$$

Por lo tanto la función de transferencia está dada por:

$$\frac{V_s(s)}{V_e(s)} = \frac{CR_2R_3s}{LC(R_2 + R_3)s^2 + (L + C(R_1R_2 + R_1R_3 + R_2R_3))s + R_1 + R_2}$$

$$e(s) = 1$$

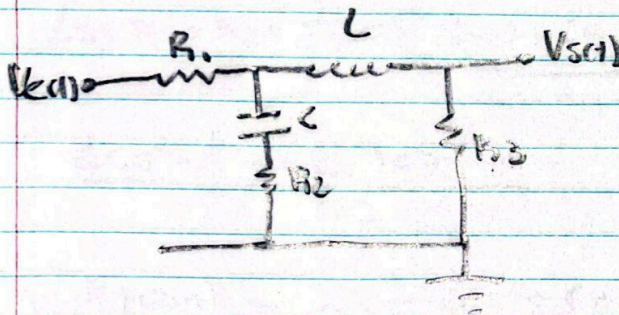


$$I_1(s) = [V(s) - \frac{1}{Cs}] \frac{1}{R_1 + R_2} + R_2 I_2(s) \frac{1}{R_1 + R_2}$$

$$I_2(s) = \left[ \frac{1}{Cs} [I_1(s) - I_2(s)] + R_2 I_1(s) - L \frac{dI_2(s)}{dt} \right] \frac{1}{R_2 + R_3}$$

$$V(s) = R_3 I_2(s)$$

$$V(s) = R_1 I_1(s) + \frac{1}{Cs} [I_1(s) - I_2(s)] + R_2 [I_1(s) - I_2(s)]$$



$$V(s) = R_1 I_1(s) + \frac{I_1(s) - I_2(s)}{Cs} + R_2 [I_1(s) - I_2(s)]$$

$$\frac{I_1(s) - I_2(s)}{Cs} + R_2 [I_1(s) - I_2(s)] = L s I_2(s) + R_3 I_2(s)$$

$$V(s) = R_3 I_2(s)$$



# Modelado de Sistemas Fis

17-abr-2026

$$V_0(t) = \left( \frac{dI_1(t)}{dt} + R_1 I_1(t) + R_2 [I_1(t) - I_2(t)] \right)$$

$$R_2 [I_1(t) - I_2(t)] = \frac{1}{C} \int I_2(t) dt + R_3 I_2(t)$$

$$V_0(t) = R_3 I_2(t)$$

Transformada de Laplace

$$V_0(s) = (s I_1(s) + R_1 I_1(s) + R_2 [I_1(s) - I_2(s) - I_2(s)])$$

$$R_2 [I_1(s) - I_2(s)] = \frac{I_2(s)}{Cs} + R_3 I_2(s)$$

$$V_0(s) = R_3 I_2(s)$$

Se despeja  $I_1(s)$  en función de  $I_2(s)$  en la segunda ecuación  
(segunda malla o igualdad de voltaje)

$$R_2 [I_1(s) - I_2(s)] = \frac{I_2(s)}{Cs} + R_3 I_2(s)$$

$$R_2 I_1(s) - R_2 I_2(s) = \frac{I_2(s)}{Cs} + R_3 I_2(s)$$

$$R_2 I_1(s) = \frac{I_2(s)}{Cs} + R_3 I_2(s) + R_2 I_2(s)$$

$$R_2 I_1(s) = \left( \frac{1}{Cs} + R_3 + R_2 \right) I_2(s)$$

$$I_1(s) = \left( \frac{1}{Cs} + R_3 + R_2 \right) \frac{I_2(s)}{R_2}$$